Supplementary: Open Questions (v8)

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Supplement to SCG-HMH: A Closed Derivative Chain for a Regenerative Nitrogen Plasma-Ring Generator, version 8 (29 October 2026; an erratum of v7, Zenodo 10.5281/zenodo.23291052;
licence CC BY 4.0). Every number here is taken from the paper; nothing is new.

Rule: nothing that lowers output or COP is booked unless it is in the published design. Each item is left out of the booked numbers; its effect is stated.

#	Open question	Effect if left out
F1	Stator temperature for fields above 5 T (design stators run at 77–85 K; the demonstrated 20 T REBCO coil ran at 20 K)	Not booked. On Earth at least 14 W per W of 20 K heat (Carnot); in space possibly passive
F3	Are the extra field's magnets on the stator (records: ReBCO stator) or on the rotor?	Stator-side taken: 0 kg extra rotor mass; whirl stays at 276,000 rpm
B2	Bearing disc size and end-face space (no disc diameter published)	None: 18.8 / 20.3 / 21.0 cm² needed at 5 / 10.94 / 13 T against 260.2 cm² per end
B3	Confirm four symmetric stators and a non-ferromagnetic rotor (centring field)	If not centring, a magnetic side load would add to the bearing load; nothing booked
B4	YBCO lift pressure (41 N/cm² at 77 K used; none published)	None on output
C1	Flow-path cross-section (410.51 kg/s through the 2 mm × 93 mm annulus would move at 52,416 m/s)	Not used; velocity reported only
M1	Protection of the rotor's N52 / Dy-NdFeB array from the 5–13 T stator field	The array's 0–4.87 % (5 T) / 0–0.721 % (13 T) contribution is at risk; nothing booked
M2	Array thickness, mass and pole count (p = 4, η = 0.900 assumed)	None booked
M3	Array carried by the Ti-Be core (case B) or pressing on the CFRP sleeve (case A)?	Case B up to +4.87 % at 5 T; case A loses output; nothing booked
S1	Fed-back share g_0 of the stator field at the booked rpm (to be measured on the bench)	At SF 2: g_0 0.10 gives 126.96 MW; ≥ 0.4727 runs to the 20 T Ic cap (4,565.46 MW). Nothing booked
S2	Stator operating temperature for the Ic cap (20 T at 77 K published)	Sets the cap field; at 77 K, 20 T

Answered and closed: B1 (bearings up-rated to 1,300 N), F2 and S3 (loss of field at speed is a fault, not a design limit).